

API REST COM SPRING BOOT

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QUEM SOU EU?

- Fausto Bernard Melo Soares
- Matemática (UFS)
- Tecnologia em Análise e Desenvolvimento de Sistemas
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O QUE VAMOS APRENDER?

- Como é feito Software nos dias de hoje?
- Conhecendo API e protocolo HTTP;
- O que é o SPRING BOOT?
- Criando uma Aplicação do Zero!
- CRUD (Create, Read, Update e Delete);
- Spring Data JPA
- DTO ou Record
- Repository
- Bean Validation
- Swagger
- Testando minha aplicação com Postman/Insomnia

REPOSITÓRIO

- Github

<https://github.com/faustosoares/todo2025>

SW NOS DIAS ATUAIS

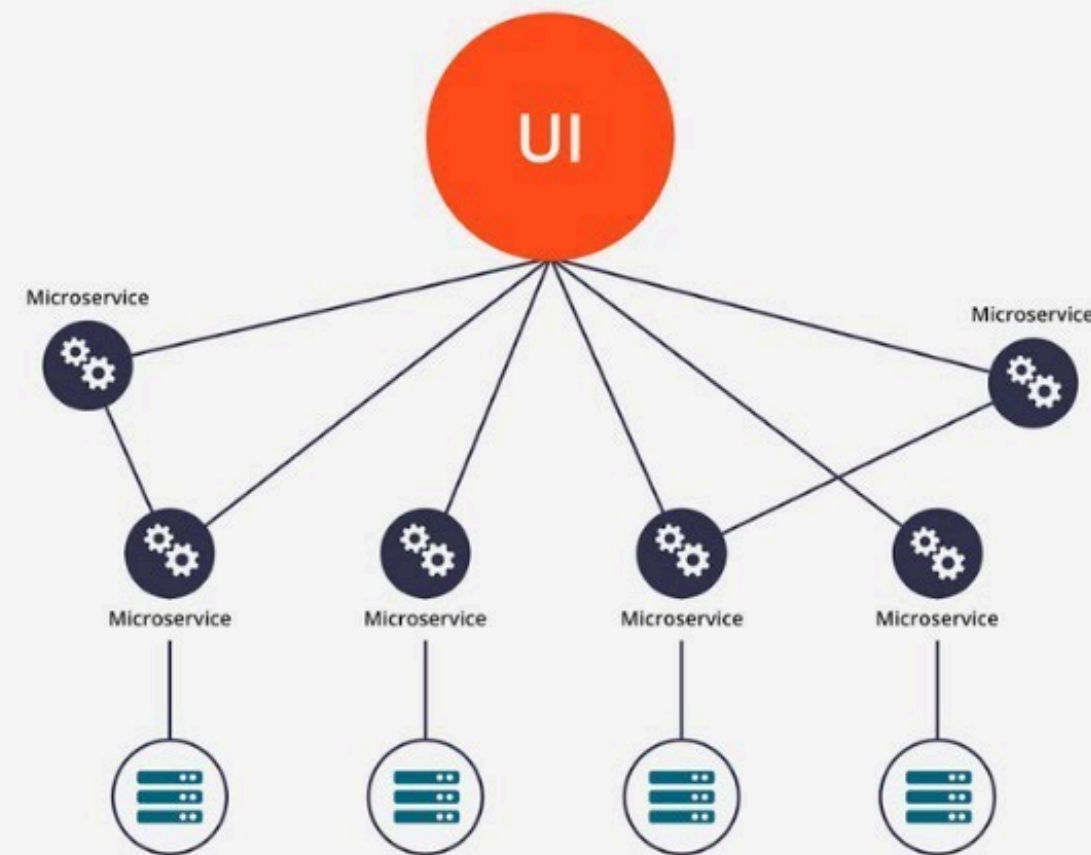
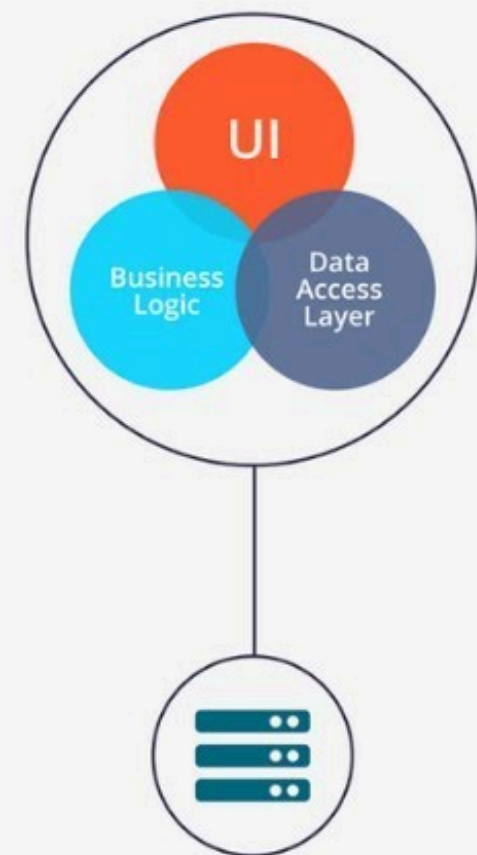
Front-End vs. Back-End Development: What's the Difference?



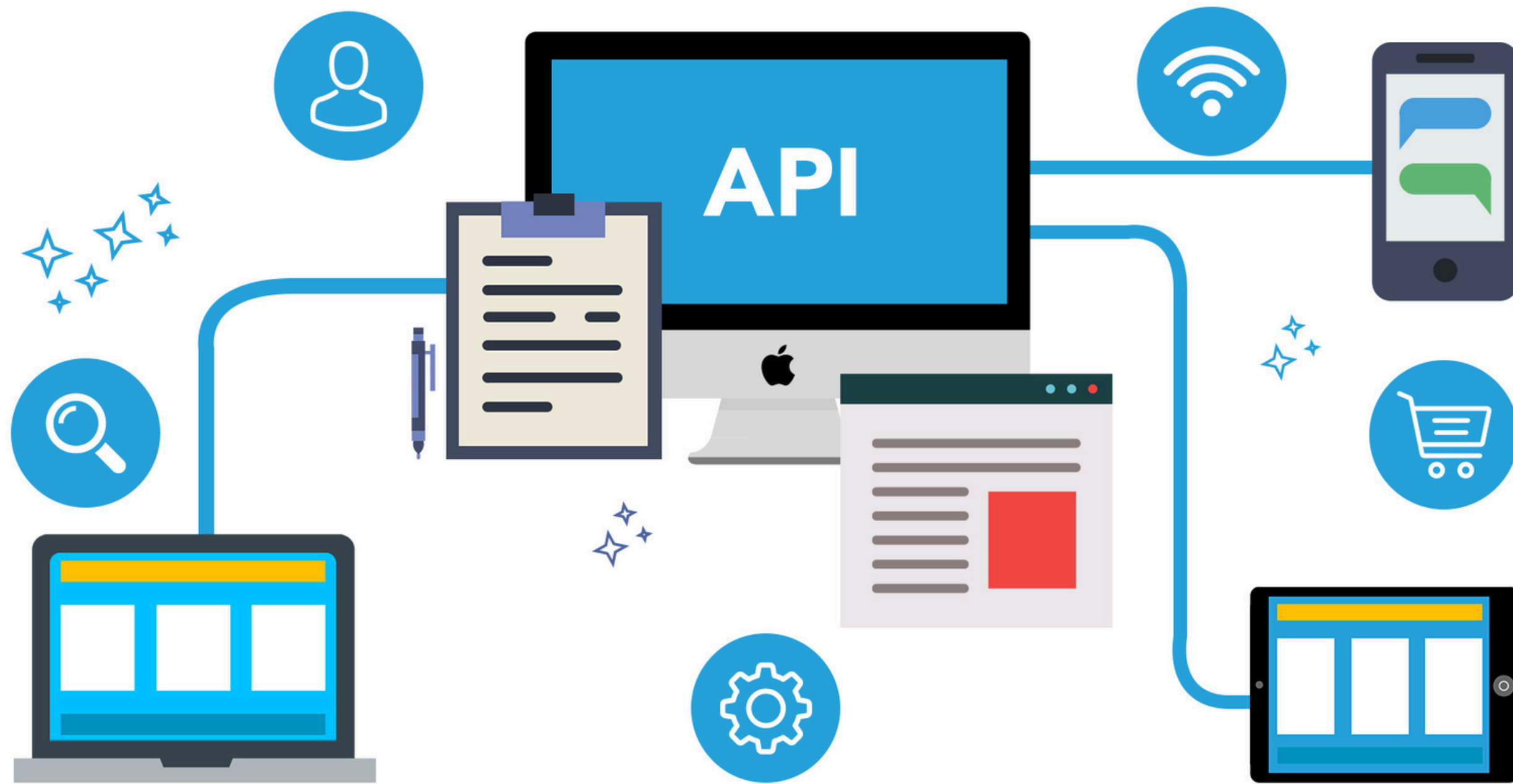
SW NOS DIAS ATUAIS

ARQUITETURA MONOLÍTICA

ARQUITETURA DE MICROSERVIÇOS



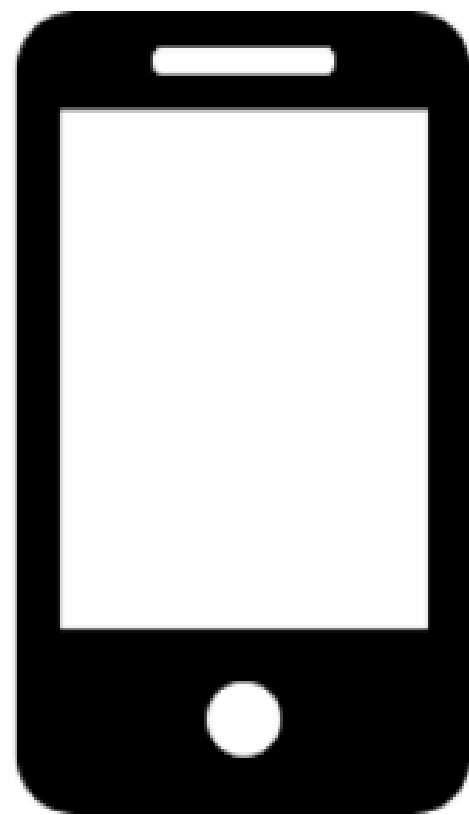
SW NOS DIAS ATUAIS



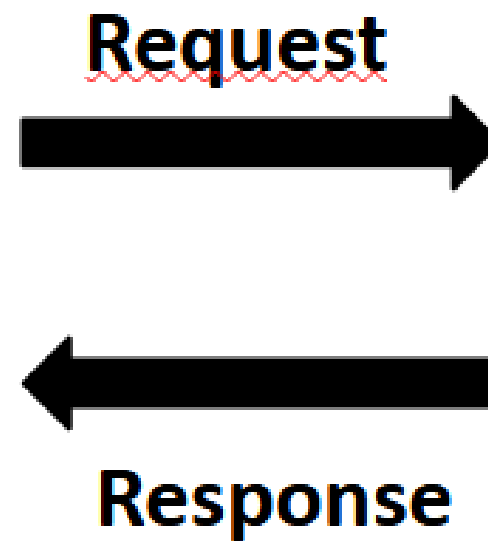
API Testing

HTTP

- Hypertext Transfer Protocol
- O protocolo HTTP possibilita a interação através de mensagens entre dois agentes



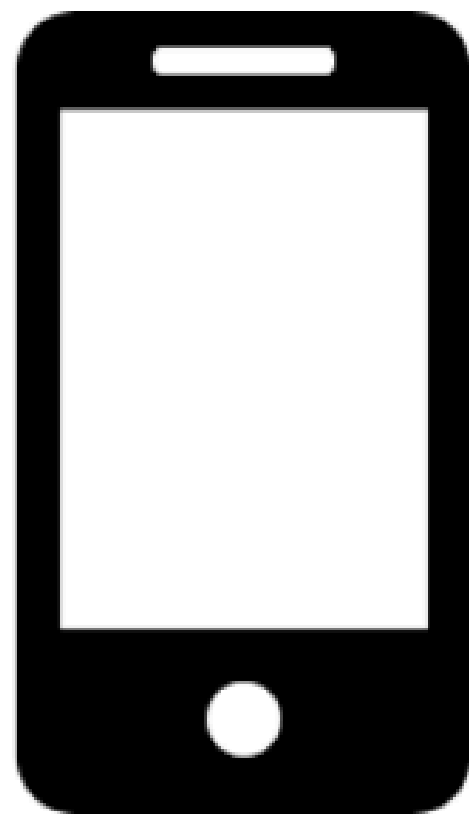
Cliente



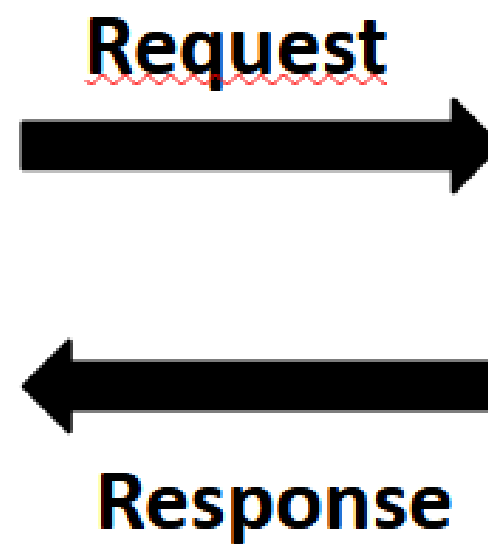
Servidor

HTTP - VERBOS

- Hypertext Transfer Protocol
- O protocolo HTTP possibilita a interação através de mensagens entre dois agentes



Cliente



Servidor

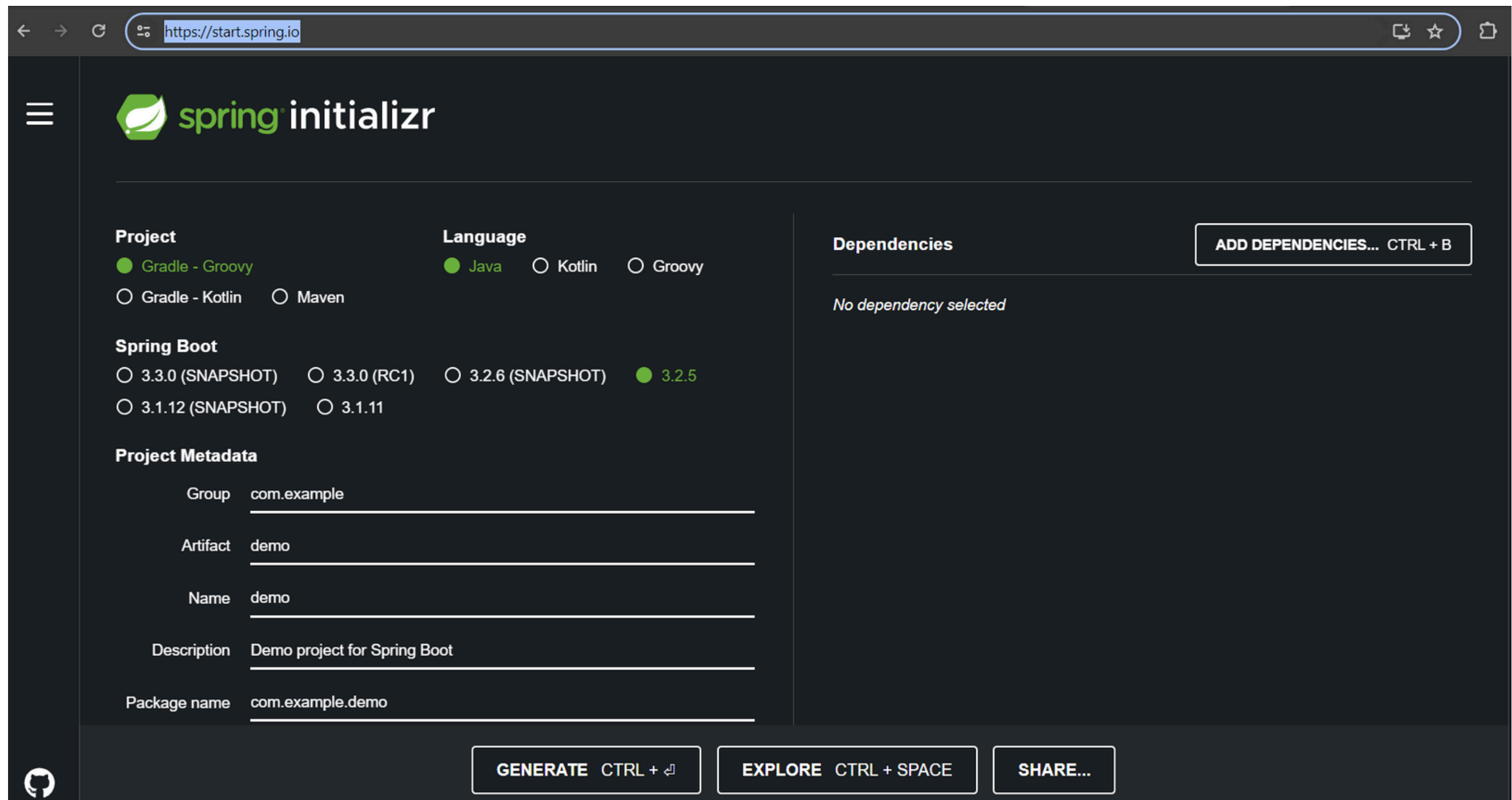
O QUE É SPRING

Framework Java que facilita o desenvolvimento de aplicações web e APIs com o ecossistema Spring. Ele foi criado para reduzir a complexidade da configuração do Spring tradicional.



CRIANDO MINHA APLICAÇÃO

<https://start.spring.io/>



The screenshot shows the Spring Initializr web application interface. The browser address bar displays <https://start.spring.io/>. The page features a dark theme with a sidebar menu on the left. The main content area is divided into three sections: Project, Language, and Dependencies. The Project section includes options for Gradle - Groovy (selected), Gradle - Kotlin, and Maven. The Language section includes options for Java (selected), Kotlin, and Groovy. The Spring Boot section includes version options: 3.3.0 (SNAPSHOT), 3.3.0 (RC1), 3.2.6 (SNAPSHOT), 3.2.5 (selected), 3.1.12 (SNAPSHOT), and 3.1.11. The Project Metadata section includes fields for Group (com.example), Artifact (demo), Name (demo), Description (Demo project for Spring Boot), and Package name (com.example.demo). The Dependencies section includes a button to ADD DEPENDENCIES... (CTRL + B) and a message indicating No dependency selected. At the bottom, there are three buttons: GENERATE (CTRL + G), EXPLORE (CTRL + SPACE), and SHARE... A GitHub logo is visible in the bottom left corner.

Project

- ☒ Gradle - Groovy
- ☐ Gradle - Kotlin
- ☐ Maven

Language

- ☒ Java
- ☐ Kotlin
- ☐ Groovy

Spring Boot

- ☐ 3.3.0 (SNAPSHOT)
- ☐ 3.3.0 (RC1)
- ☐ 3.2.6 (SNAPSHOT)
- ☒ 3.2.5
- ☐ 3.1.12 (SNAPSHOT)
- ☐ 3.1.11

Project Metadata

Group:

Artifact:

Name:

Description:

Package name:

Dependencies

[ADD DEPENDENCIES... CTRL + B](#)

No dependency selected

[GENERATE CTRL + G](#) [EXPLORE CTRL + SPACE](#) [SHARE...](#)

CRIANDO MINHA APLICAÇÃO

start.spring.io

Project

☐ Gradle - Groovy

☐ Gradle - Kotlin

☒ Maven

Spring Boot

☐ 3.3.0 (SNAPSHOT)

☐ 3.3.0 (RC1)

☐ 3.2.6 (SNAPSHOT)

☒ 3.2.5

☐ 3.1.12 (SNAPSHOT)

☐ 3.1.11

Project Metadata

Group

br.com.estacio

Artifact

todo

Name

ToDo

Description

API ToDo List

Package name

br.com.estacio.todo

Packaging

☒ Jar

☐ War

Java

☐ 22

☒ 21

☐ 17

Language

☒ Java

☐ Kotlin

☐ Groovy

Dependencies

ADD DEPENDENCIES... CTRL + B

Spring Web

WEB

Build web, including RESTful, applications using Spring MVC. Uses Apache Tomcat as the default embedded container.

Spring Boot DevTools

DEVELOPER TOOLS

Provides fast application restarts, LiveReload, and configurations for enhanced development experience.

Spring Boot Actuator

OPS

Supports built in (or custom) endpoints that let you monitor and manage your application - such as application health, metrics, sessions, etc.

GENERATE CTRL + ↵

EXPLORE CTRL + SPACE

SHARE...

CONHECENDO ARQUIVOS GERADOS

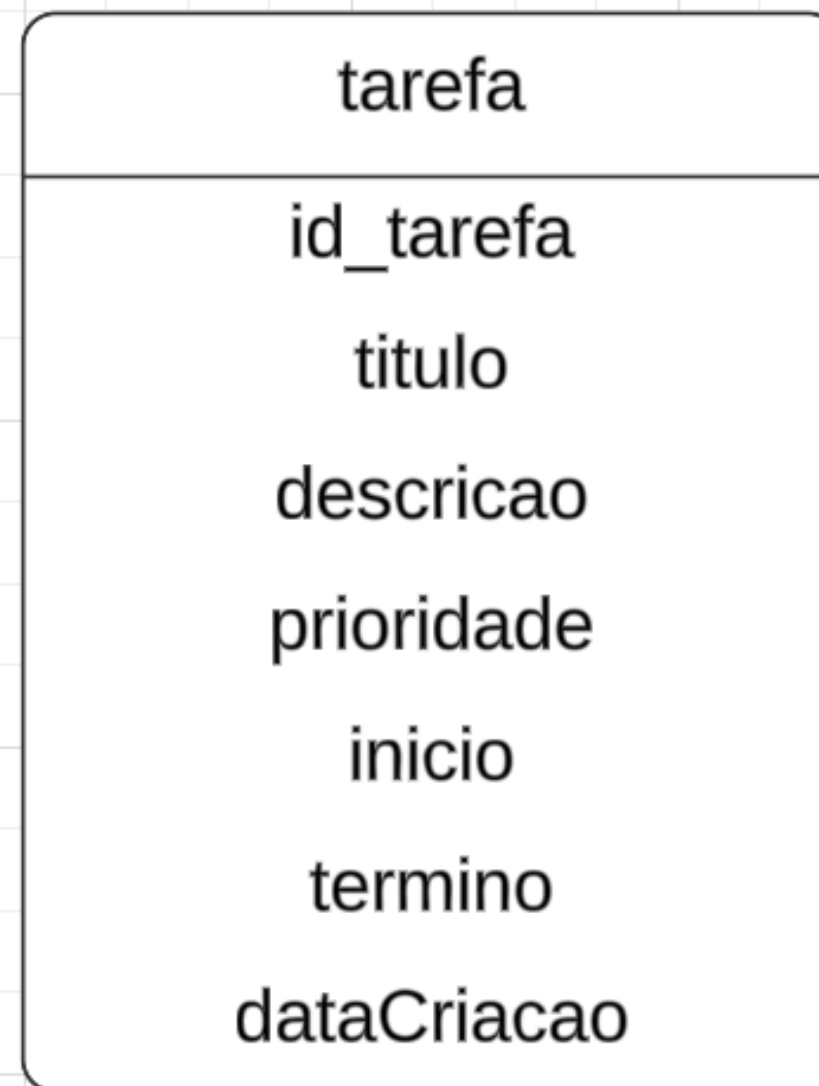
- src: nossos fontes
- resources: configurações
- test: classes de teste
- Maven dependencies: libs associadas ao projeto
- pom.xml: arquivo de configuração do Maven



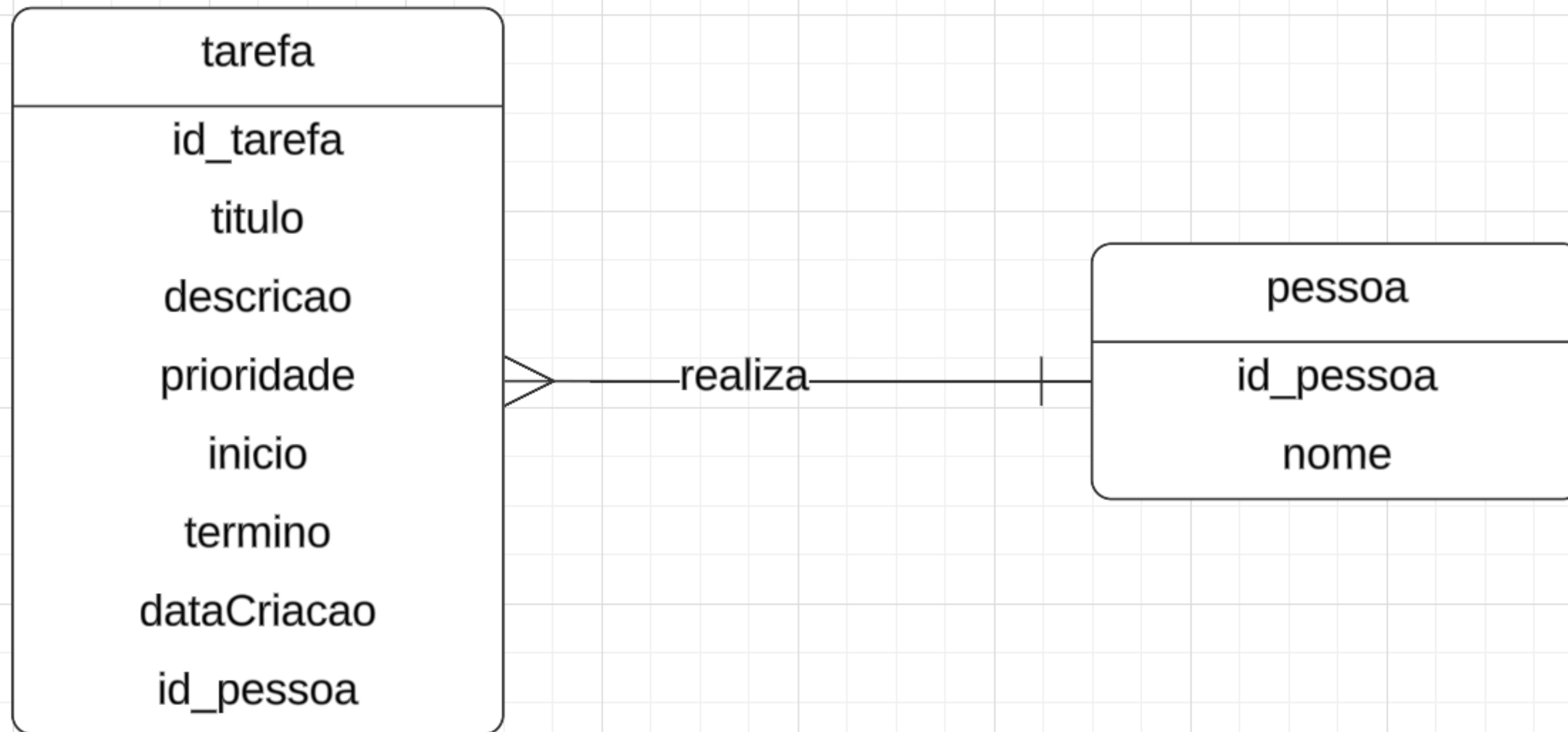
TODOLIST

- Implemente um sistema de lista conforme slides a seguir

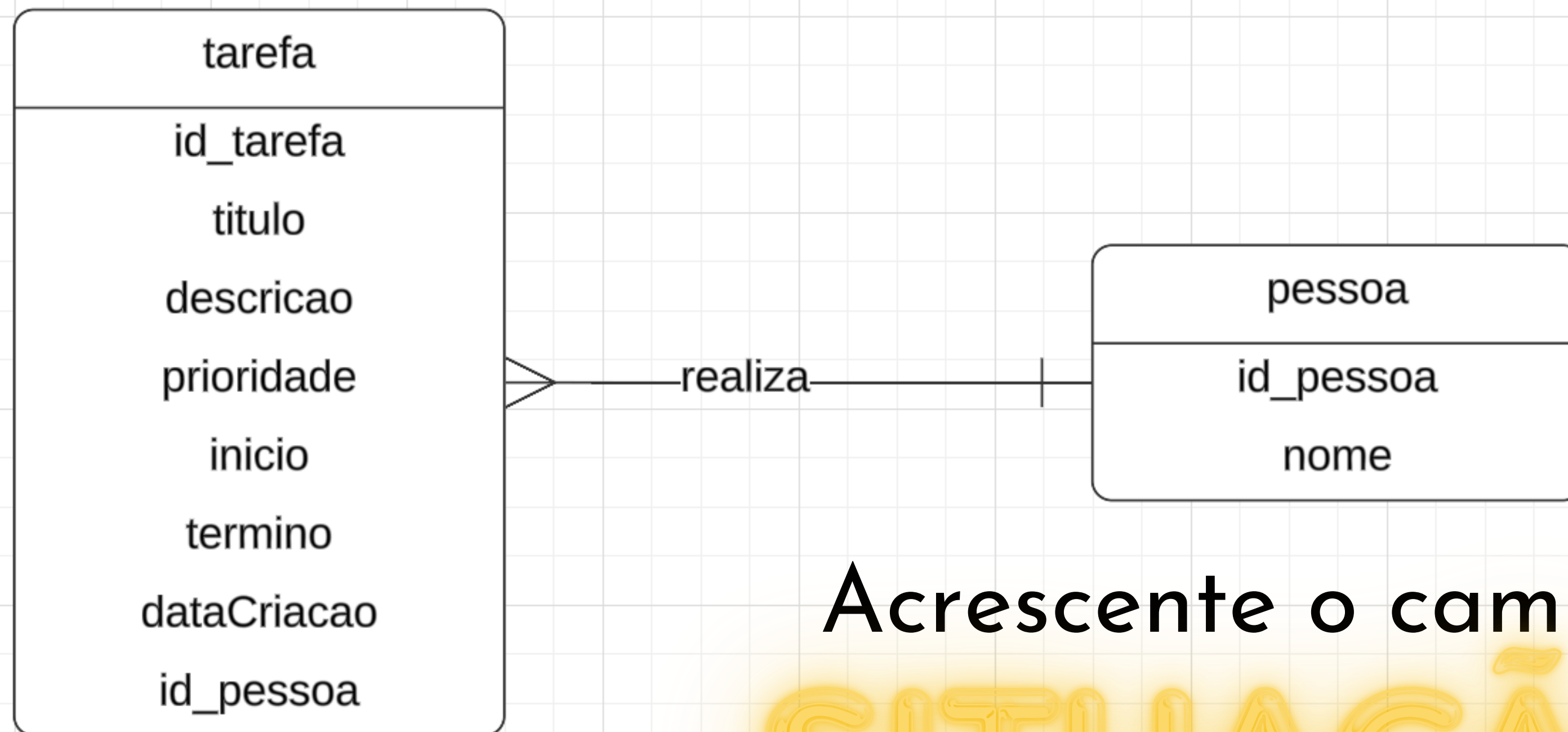
MODELANDO NOSSO PROBLEMA



MODELANDO NOSSO PROBLEMA



MODELANDO NOSSO PROBLEMA



Acréscimo o campo:

SITUAÇÃO